ELEC40001-B - Mathematics: Principles and Engineering Applications - Solutions 2020

1. (a) A periodic function is defined as

$$h(t) = \begin{cases} 1-t & \text{for} \quad 0 \le t < 1 \\ 2t-1 & \text{for} \quad 1 \le t < 2 \end{cases}$$

and $h(t+2) = h(t)$ for all t. A Fourier series is obtained for this function [*You do not need to do this!*]. What are the values of the Fourier series at $t = 0$ and $t = 3$?

SOLUTION

Take the average of the two sides at a discontinuity. On $[-1, 0]$ we have $h(t) = 2t + 3$, as a sketch would show, so at

$$t = 0 : \qquad \frac{1}{2}\left([1-0] + [2(0)+3]\right) = 2\,.$$

Then at $t = 3$, given periodicity, same as at

$$t = 1 : \qquad \frac{1}{2}\left([1-1] + [2(1)-1]\right) = \frac{1}{2}\,.$$

Mark: 1 for trying, 2 for one correct, 3 for both correct

(b) A function $f(t)$ is defined as

$$f(t) = \begin{cases} 0 & \text{for} \quad 0 \le t < \frac{1}{2} \\ t - \frac{1}{2} & \text{for} \quad \frac{1}{2} \le t < 1\,. \end{cases}$$

Let $g_1(t)$ and $g_2(t)$ be the even and odd extensions, respectively, of $f(t)$, both with period $T = 2$. Sketch g_1 and g_2, for $-3 \le t \le 3$.

SOLUTION Sketch at end.

Mark: 1 for trying, 2 for one correct, 3 for both correct

(c) Calculate the Fourier Transform of the function

$$f(t) = \begin{cases} \cos(t)\,, & \text{for} \quad -\pi \le t \le \pi\,, \\ 0\,, & \text{otherwise} \end{cases}$$

in terms of $\sin(\pi\omega)$.

SOLUTION

Use the definition of Fourier transform,

$$F(\omega) \;=\; \int_{-\infty}^{+\infty} f(t)e^{-i\omega t}\,dt \;=\; \int_{-\pi}^{\pi} \cos(t)e^{-i\omega t}\,dt \;=\; \int_{-\pi}^{\pi} \frac{e^{it}+e^{-it}}{2}e^{-i\omega t}\,dt \;=\; \frac{1}{2}\int_{-\pi}^{\pi} e^{(1-\omega)ti} + e^{-(1+\omega)ti}\,dt$$

$$= \frac{1}{2}\left[\frac{e^{(1-\omega)ti}}{(1-\omega)i} + \frac{e^{-(1+\omega)ti}}{-(1+\omega)i}\right]_{-\pi}^{\pi} = \frac{1}{2i}\left[\frac{e^{(1-\omega)\pi i}}{1-\omega} - \frac{e^{-(1-\omega)\pi i}}{1-\omega} - \frac{e^{-(1+\omega)\pi i}}{1+\omega} + \frac{e^{(1+\omega)\pi i}}{1+\omega}\right]$$

and as $e^{\pm i\pi} = -1$, this reduces to

$$F(\omega) = -\frac{1}{2i}\left[\frac{e^{-\omega\pi i}}{1-\omega} - \frac{e^{\omega\pi i}}{1-\omega} - \frac{e^{-\omega\pi i}}{1+\omega} + \frac{e^{\omega\pi i}}{1+\omega}\right] = -\frac{1}{2i}\left(e^{\omega\pi i} - e^{-\omega\pi i}\right)\left(\frac{1}{\omega+1} + \frac{1}{\omega-1}\right) = \frac{2\omega}{1-\omega^2}\sin(\pi\omega)$$

as required.

Mark: 1 for trying, 2 for halfway, 4 for completely correct

(d) Obtain the cartesian equation of the plane through the points $A(1,2,-3)$, $B(0,1,2)$ and $C(3,4,-5)$.

SOLUTION

Two vectors in the plane are $\overrightarrow{BA} = (1,1,-5)$ and $\overrightarrow{AC} = (2,2,-2)$ so a normal vector can be taken as $\underline{\mathbf{n}} = \overrightarrow{BA} \times \overrightarrow{AC} = (8,-8,0)$ and the plane has equation

$$(x,y,z)\cdot(8,-8,0) = (1,2,-3)\cdot(8-8,0) \Rightarrow x - y = -1\,.$$

Mark: 2 for small errors, 3 for correct

(e) Find the distance from the point $P(1,-2,2)$ to the plane with cartesian equation $-2x+2y+z=5$. Find the point Q on the plane, which is nearest to P.

SOLUTION

The line through P and perpendicular to the plane is

$$\underline{\mathbf{r}} = (x,y,z) = (1,-2,2) + \lambda(-2,2,1)$$

which we substitute into the equation of the plane to find

$$-2(1-2\lambda) + 2(-2+2\lambda) + 1(2+\lambda) = 5 \Rightarrow \lambda = 1$$

giving the point of intersection of line and plane as $Q = (1,-2,2) + 1(-2,2,1) = (-1,0,3)$. Hence the distance is $|\overrightarrow{PQ}| = |(-2,2,1)| = \sqrt{4+4+1} = 3$

Mark: 3 for either point or distance, using projection/intersection, 4 for both

(f) Use Gaussian elimination to find the inverse of the matrix $\begin{pmatrix} 3 & -1 & 2 \\ -4 & 1 & -2 \\ 1 & 1 & 1 \end{pmatrix}$.

SOLUTION

Set up $(A : I)$ and use Gauss-Jordan elimination:

$$\left(\begin{array}{ccc|ccc} 3 & -1 & 2 & 1 & 0 & 0 \\ -4 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \underset{R_3 \leftrightarrow R_1}{\sim} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 0 & 1 \\ -4 & 1 & -2 & 0 & 1 & 0 \\ 3 & -1 & 2 & 1 & 0 & 0 \end{array}\right) \underset{\substack{R_2+4R_1 \\ R_3-3R_1}}{\sim} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 5 & 2 & 0 & 1 & 4 \\ 0 & -4 & -1 & 1 & 0 & -3 \end{array}\right)$$

To avoid fractions too early, employ a trick and get a pivot $a_{22} = 1$:

$$\underset{R_2+R_3}{\sim} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & -4 & -1 & 1 & 0 & -3 \end{array}\right) \underset{\substack{R_1-R_2 \\ R_3+4R_2}}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 3 & 5 & 4 & 1 \end{array}\right) \underset{\frac{1}{3}R_3}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & \frac{5}{3} & \frac{4}{3} & \frac{1}{3} \end{array}\right)$$

$$\underset{R_2-R_3}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & -\frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 1 & \frac{5}{3} & \frac{4}{3} & \frac{1}{3} \end{array}\right) \text{ Hence the inverse is } \left(\begin{array}{ccc} -1 & -1 & 0 \\ -\frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \\ \frac{5}{3} & \frac{4}{3} & \frac{1}{3} \end{array}\right).$$

Mark: 2 for lots of errors, 3 for only a few errors, 4 for correct

(g) Given

$$A = \begin{pmatrix} -1 & 3 & 2 \\ 1 & 1 & 1 \\ 1 & -4 & -2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 1 & -2 \\ 2 & -2 & -1 \\ -3 & 1 & 2 \end{pmatrix},$$

obtain $\det(A)$, $\det(B)$ and $\det(AB)$.

SOLUTION

Expanding by the first row, $\det(A) = -1(-2+4) - 3(-2-1) + 2(-4-1) = -2 + 9 - 10 = -3$

Similarly, $\det(B) = 1(-4+1) - 1(4-3) - 2(2-6) = -3 - 1 + 8 = 4$.
So $\det(AB) = \det(A)\det(B) = -3(4) = -12$.

Mark: 1 each

(h) Find the stationary points of the function

$$f(x, y) = y(x+2)^2 + y^2 - 4y\,.$$

SOLUTION

Find the first derivatives and set them equal to zero:

$$(i)\ f_x = 2y(x+2) = 0\,, \qquad (ii)\ f_y = (x+2)^2 + 2y - 4 = 0$$

From (i) $\Longrightarrow x = -2$ or $y = 0$ which we substitute into (ii):

$$x = -2 \Longrightarrow 2y - 4 = 0 \Longrightarrow y = 2$$

so $P_1(-2, 2)$ is a stationary point;

$$y = 0 \Longrightarrow (x+2)^2 = 4 \Longrightarrow x = 0, -4$$

so $P_2(0, 0)$ and $P_3(-4, 0)$ are stationary points.

Mark: 2 for small errors, 3 for correct

(i) Classify the stationary points of the function

$$f(x, y) = y(x + 2)^2 + y^2 - 4y\,,$$

which you found in the previous question.

SOLUTION

To classify, obtain the Hessian determinant:
$D = f_{xx}f_{yy} - (f_{xy})^2 = (2y)(2) - [2(x + 2)]^2 = 4[y - (x + 2)^2]$.

At $(0, 0)$ and $(-4, 0)$ we have $D = 4(-4) < 0$ so $P_{2,3}$ are saddle points.

At $(-2, 2)$, we have $D = 4(2) > 0$ so P_1 is an extremum, and as $f_{yy} = 2 > 0$ everywhere, we see that P_1 is a minimum.

Mark: 2 for small errors, 3 for correct

2. (a) Obtain the first five non-zero terms of the Fourier Series of $g_1(t)$, the period-2 even extension of $f(t)$ from question 1(b):

$$f(t) = \begin{cases} 0 & \text{for} \quad 0 \le x < \frac{1}{2} \\ t - \frac{1}{2} & \text{for} \quad \frac{1}{2} \le x < 1 \, . \end{cases}$$

SOLUTION

The function is an even extension, so all b_n are zero, and $a_n = \dfrac{2}{T}\displaystyle\int_{-T/2}^{T/2} f(t)\cos(2n\pi t/T)\; dt$ so with period $T = 2$ we have $a_n = \displaystyle\int_{-1}^{1} f(t)\cos(n\pi t)\; dt$. As f and cosine are even, the integrand is even, and the half-range series gives us

$$a_n = 2\int_0^1 f(t)\cos(n\pi t)\; dt = 2\int_{1/2}^1 \left(t - \frac{1}{2}\right)\cos(n\pi t)\; dt \, .$$

First obtain the constant term:

$$a_0 = 2\int_{1/2}^1 \left(t - \frac{1}{2}\right)\; dt \;=\; 2\left[\frac{t^2}{2} - \frac{1}{2}t\right]_{1/2}^1 = \frac{1}{4}$$

For $n \ge 1$, we integrate by parts:

$$= 2\left[\left(t - \frac{1}{2}\right)\frac{\sin(n\pi t)}{n\pi} - \int_{1/2}^1 \frac{\sin(n\pi t)}{n\pi}\; dt\right] = 2\left[\left(t - \frac{1}{2}\right)\frac{\sin(n\pi t)}{n\pi} + \frac{\cos(n\pi t)}{(n\pi)^2}\right]_{1/2}^1$$

$$= 2\left[0 - 0 + \frac{\cos(n\pi)}{(n\pi)^2} - \frac{\cos(n\pi/2)}{(n\pi)^2}\right] = \frac{2}{(n\pi)^2}\left[\cos(n\pi) - \cos\left(\frac{n\pi}{2}\right)\right] \qquad \textbf{(**)}$$

At this point care needs to be taken as the second cosine term vanishes for odd n, and there are alternating signs, depending on whether n is even or odd. So

$a_1 = \dfrac{2}{\pi^2}\left(\cos\pi - \cos(\pi/2)\right) = \dfrac{2}{\pi^2}(-1 - 0) = -\dfrac{2}{\pi^2}$, and $a_2 = \dfrac{2}{4\pi^2}\left(\cos 2\pi - \cos(\pi)\right) = \dfrac{1}{\pi^2}$.

Similarly, $a_3 = \dfrac{2}{9\pi^2}\left(\cos 3\pi - \cos(3\pi/2)\right) = -\dfrac{2}{9\pi^2}$, and $a_4 = \dfrac{2}{16\pi^2}\left(\cos 4\pi - \cos(2\pi)\right) = 0$

The pattern is clear, and $a_5 = -\dfrac{2}{(5\pi)^2}$ and $a_6 = \dfrac{4}{(6\pi)^2}$, giving five non-zero terms.

Finally, the cosine-series is $g_1(t) = \dfrac{1}{2}a_0 + \displaystyle\sum_{n=1}^{\infty} a_n \cos(n\pi t)$ and the first five non-zero terms are

$$\frac{1}{4} - \frac{2}{\pi^2}\cos(\pi t) + \frac{1}{\pi^2}\cos(2\pi t) - \frac{2}{9\pi^2}\cos(3\pi t) - \frac{2}{25\pi^2}\cos(5\pi t) + \frac{1}{9\pi^2}\cos(6\pi t)$$

Mark: 5 marks up to (**), 8 for completely correct

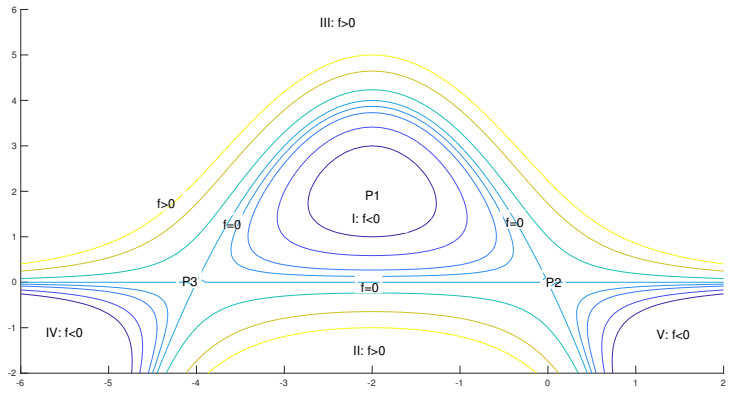
(b) In question 1(h) and (i), you found and classified the stationary points of $f(x,y) = y(x+2)^2+y^2-4y$. Now use these to sketch the contours of the function.

SOLUTION

To sketch the contours, first factorize: $f(x,y) = y[(x+2)^2 + y - 4]$ and setting

$$f(x,y) = y\left[(x+2)^2 + y - 4\right] = 0 \Rightarrow y = 0 \quad \text{or} \quad (x+2)^2 + y - 4 = 0$$

we see that the function takes the value zero on the $x-$axis and the parabola $y = -(x+2)^2+4$. These contours intersect at the saddle points $P_2(0,0)$ and $P_3(-4,0)$. The minimum at $P_1(-2,2)$ is enclosed by the axis and parabola, in region I in the second quadrant, and contours in this region are all $f < 0$. Moving vertically down from P_1 and crossing the zero-contour into the third quadrant, we obtain $f > 0$ and continuing in this direction, we have $f \to \infty$ in region II, including parts of the third and fourth quadrants, in the figure below. Similarly moving vertically up from P_1 we have region III in the first and second quadrants, where again $f > 0$ and $f \to \infty$. Crossing the zero-contour from region II into region IV in the third quadrant, we move from $f > 0$ to $f < 0$, giving negative contours in region IV and similarly region V in the fourth, with $f \to -\infty$. Finally, for large x or y, f changes faster so contours are closer together further away from the stationary points. All of the above should be clearly visible in the solution sketch.



Mark: 3 for correct sketch but no explanation, 4 with minimal explanation, 5 for more than one comment

(c) Given the matrix

$$A = \begin{pmatrix} 2 & -2 & 3 \\ -1 & 1 & -2 \\ 3 & 2 & 4 \end{pmatrix};$$

(i) Use Gaussian elimination to show that A cannot be written in the form $A = LU$ where L is lower triangular and U upper triangular. Hence, identify the permutation matrix P needed so that $LU-$decomposition can be done on PA;

SOLUTION

Using $a_{11} = 2$ as pivot, the first two elimination operations are $R_2 + \frac{1}{2}R_1$ and $R_3 - \frac{3}{2}R_1$ so that

$$A = \begin{pmatrix} 2 & -2 & 3 \\ -1 & 1 & -2 \\ 3 & 2 & 4 \end{pmatrix} \sim \begin{pmatrix} 2 & -2 & 3 \\ 0 & 0 & -\frac{1}{2} \\ 0 & 5 & -\frac{1}{2} \end{pmatrix}$$

This cannot be transformed into an upper triangular matrix without a row swap, so $LU-$decomposition, depeding solely on elimination, is not possible. The permutation matrix needed to effect the row swap $R_3 \leftrightarrow R_2$ to enable $LU-$decomposition is

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

An alternative would be $R_1 \leftrightarrow R_3$, equally acceptable.

(ii) Use Gaussian elimination to transform PA into the upper triangular U and write down the elimination matrices needed for this;

SOLUTION

With the given permuation matrix we proceed

$$PA = \begin{pmatrix} 2 & -2 & 3 \\ 3 & 2 & 4 \\ -1 & 1 & -2 \end{pmatrix} \underset{\substack{R_2 - \frac{3}{2}R_1 \\ R_3 + \frac{1}{2}R_1}}{\sim} \begin{pmatrix} 2 & -2 & 3 \\ 0 & 5 & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} \end{pmatrix} = U$$

and the elimination matrices corresponding to the row operations are

$$E_{21} = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ and } E_{31} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{pmatrix}$$

(iii) Write down the inverses of the elimination matrices;

SOLUTION

The inverses of the elimination matrices are

$$E_{21}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ and } E_{31}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \end{pmatrix}$$

(iv) Hence, or otherwise, find the corresponding lower triangular L so that $PA = LU$.

SOLUTION

Hence, $L = E_{21}^{-1}E_{31}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \end{pmatrix}$.

Mark: 3 for (i), 3 for (ii), 2 for the rest.

(d) Using Laplace transforms, or otherwise, solve

$$\frac{d^2x}{dt^2} + 4x = \begin{cases} 4 & \text{for} \quad 0 \le t \le 1 \\ 0 & \text{for} \quad t > 1 \end{cases}$$

and $x(0) = 0, \quad x'(0) = 1$, writing out the solution explicitly for the intervals $0 < t < 1$ and $t > 1$.

SOLUTION

Taking Laplace transforms of the ODE we get

$$s^2\bar{x} - sx(0) - x'(0) + 4\bar{x} = 4\mathcal{L}[H(t) - H(t-1)] = \frac{4}{s} - \frac{4e^{-s}}{s}$$

and using the initial conditions we have

$$(s^2+4)\bar{x} = \frac{4}{s} - \frac{4e^{-s}}{s} + 1 \Longrightarrow \bar{x} = \frac{1}{s^2+4} + \frac{4}{s(s^2+4)}\left(1 - e^{-s}\right)$$

Using partial fractions,

$$\frac{4}{s(s^2+4)} = \frac{1}{s} - \frac{s}{s^2+4}\,,$$

and so

$$\bar{x} = \frac{1}{s^2+4} + \frac{1}{s} - \frac{s}{s^2+4} - \left(\frac{1}{s} - \frac{s}{s^2+4}\right)e^{-s}$$

and using inverse transforms and the second shift theorem, we get

$$x(t) = \frac{1}{2}\sin(2t) + 1 - \cos(2t) - H(t-1)\left[1 - \cos(2t)\right]_{t\to t-1}$$

$$= 1 + \frac{1}{2}\sin(2t) - \cos(2t) + H(t-1)[\cos 2(t-1) - 1]\,.$$

Finally,

$$x(t) = \begin{cases} 1 + \frac{1}{2}\sin(2t) - \cos(2t)\,, & \text{for} \quad 0 < t < 1\,, \\ \frac{1}{2}\sin(2t) - \cos(2t) + \cos 2(t-1)\,, & \text{for} \quad t > 1\,. \end{cases}$$

Mark: 4 for correct $\bar{x}(s)$, 3 for correct $x(t)$.

(e) Given the linear system of equations $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$ where

$$A = \begin{pmatrix} 1 & 1 & 1 & 4 \\ 1 & 0 & 2 & 1 \\ 1 & -3 & 5 & -8 \end{pmatrix},$$

(i) For $\underline{\mathbf{b}} = \begin{pmatrix} 1 \\ -1 \\ \alpha \end{pmatrix}$, obtain the echelon form of the augmented coefficient matrix.

SOLUTION $(A : \underline{\mathbf{b}}) =$

$$\begin{pmatrix} 1 & 1 & 1 & 4 & 1 \\ 1 & 0 & 2 & 1 & -1 \\ 1 & -3 & 5 & -8 & \alpha \end{pmatrix} \underset{\substack{R_2-R_1 \\ R_3-R_1}}{\sim} \begin{pmatrix} 1 & 1 & 1 & 4 & 1 \\ 0 & -1 & 1 & -3 & -2 \\ 0 & -4 & 4 & -12 & \alpha - 1 \end{pmatrix} \underset{R_3 - 4R_2}{\sim} \begin{pmatrix} 1 & 1 & 1 & 4 & 1 \\ 0 & -1 & 1 & -3 & -2 \\ 0 & 0 & 0 & 0 & \alpha + 7 \end{pmatrix}$$

(ii) Identify free and pivot columns, and free and pivot variables; state the rank of A.

SOLUTION

Pivots are $a_{11} = 1$ and $a_{22} = -1$, so columns 1 and 2 are pivot columns, while columns 3 and 4 are free columns. In $\underline{\mathbf{x}} = (x_1, x_2, x_3, x_4)^T$, this means $x_{1,2}$ are pivot variables and $x_{3,4}$ are free variables; rank$(A) = 2$.

(iii) Use the free variables to find the special solutions of $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$. What does each special solution say about the columns of A?

SOLUTION

To find the special solutions of $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$ we let the free variables be 1,0 and vice versa. Row 2 of the echelon form implies $-x_2 + x_3 - 3x_4 = 0$ while row 1 has $x_1 + x_2 + x_3 + 4x_4 = 0$ Then, if $x_3 = 1$ and $x_4 = 0$, then row 2 has $-x_2 + 1 = 0 \Rightarrow x_2 = 1$ and row 1 has $x_1 + 1 + 1 = 0 \Rightarrow x_1 = -2$ and the first special solution is

$$\underline{\mathbf{x}}_1^s = \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \end{pmatrix}.$$

Taking $A\underline{\mathbf{x}}$ as a linear combination of the columns c_i of A, we have $A\underline{\mathbf{x}}_1^s = \underline{\mathbf{0}}$ giving $-2c_1 + c_2 + c_3 = \underline{\mathbf{0}}$, i.e. the second and third column of A add up to twice the first column, easily confirmed, but not easy to spot.

Now let $x_3 = 0$ and $x_4 = 1$, then row 2 has $-x_2 - 3 = 0 \Rightarrow x_2 = -3$ and row 1 has $x_1 - 3 + 4 = 0 \Rightarrow x_1 = -1$ and the second special solution is

$$\underline{\mathbf{x}}_2^s = \begin{pmatrix} -1 \\ -3 \\ 0 \\ 1 \end{pmatrix}.$$

This gives the arrangement of columns $-c_1 - 3c_2 + c_4 = \underline{\mathbf{0}}$, i.e. the first column plus three times the second column is equal to the fourth column; again: not easy to spot, but easily confirmed.

(iv) Give the nullspace and column space of A.

SOLUTION

The nullspace is all linear combinations of the special solutions:

$$N(A) = \left\{ p \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \end{pmatrix} + q \begin{pmatrix} -1 \\ -3 \\ 0 \\ 1 \end{pmatrix} \mid p, q \in \mathbb{R} \right\},$$

a subspace of $\mathbb{R}^4$.
The column space is all linear combinations of the pivot columns of A:

$$\mathrm{col}(A) = \left\{ p \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + q \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix} \mid p, q \in \mathbb{R} \right\},$$

a subspace of $\mathbb{R}^3$.

(v) Find a condition on α to guarantee solvability of $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$. Hence, obtain the general solution.

SOLUTION

Solvability requires that row 3 of $(A : \underline{\mathbf{b}})$ after elimination is not impossible. Row 3 of the echelon form implies that $0x_1 + 0x_2 + 0x_3 + 0x_4 = 0 = \alpha + 7$, so we need $\alpha = -7$.

We now need one particular solution to $A\underline{\mathbf{x}} = \underline{\mathbf{b}} = (1, -1, 5)^T$; this is obtained by setting all the free variables equal to zero. Row 2 of the echelon form implies $-x_2 + x_3 - 3x_4 = -2$ while row 1 has $x_1 + x_2 + x_3 + 4x_4 = 1$. Setting $x_{3,4} = 0$ we get $x_2 = 2$ and $x_1 = -1$ and the particular solution is

$$\underline{\mathbf{x}}_p = \begin{pmatrix} -1 \\ 2 \\ 0 \\ 0 \end{pmatrix},$$

giving the general solution

$$\underline{\mathbf{x}} = \begin{pmatrix} -1 \\ 2 \\ 0 \\ 0 \end{pmatrix} + p \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \end{pmatrix} + q \begin{pmatrix} -1 \\ -3 \\ 0 \\ 1 \end{pmatrix}$$

where p, q are any real numbers.

Mark: 3 for (i,ii), 3 for (iii), 2 (iv), 2 for (v)

(f) The following solution of a first-order ordinary differential equation (ODE) is incorrect.

The first order ODE

$$\left(x^2 + \frac{1}{y} + e^y\right)\frac{dy}{dx} + 2xy + \frac{1}{x} = 0 \tag{1}$$

can be seen to be exact by performing the standard check. We write the solution as $f(x, y) = constant$, and compute

$$\frac{\partial f}{\partial x} = 2xy + \frac{1}{x} \quad \Rightarrow \quad f = x^2 y + \ln(x) + g(y). \tag{2}$$

From this and the original equation we have

$$\frac{\partial f}{\partial y} = x^2 + g' = x^2 + \frac{1}{y} + e^y \tag{3}$$

which gives that

$$g = \ln(y) + e^y. \tag{4}$$

Hence the solution is

$$x^2 y - e^y + \ln\left(\frac{x}{y}\right) = constant. \tag{5}$$

Which was the *first* error made in this calculation? Eliminate the error and obtain the correct solution.

SOLUTION

The approach is different from that done in lectures, but equivalent. For

$$P(x, y) + Q(x, y)\frac{dy}{dx} = 0$$

identify

$$\frac{\partial f}{\partial x} = P(x, y) = 2xy + \frac{1}{x} \Rightarrow f(x, y) = \int P(x, y)\, dx + g(y) = x^2 y + \ln x + g(y)$$

where $g(y)$ is an arbitrary function. As

$$\frac{\partial f}{\partial y} = Q(x, y)$$

we get

$$\frac{\partial f}{\partial y} \;=\; \frac{\partial}{\partial y}\left(x^2 y + \ln x + g(y)\right) = x^2 + \frac{dg}{dy}$$

which must be equal to $Q(x, y) = x^2 + \dfrac{1}{y} + e^y$ and we correctly calculate

$$\frac{dg}{dy} = \frac{1}{y} + e^y \Rightarrow g(y) = \ln y + e^y\,.$$

Hence

$$f(x, y) = x^2 y + \ln x + \ln y + e^y$$

and the error is in the last step, equation (5), where $g(y)$ was subtracted instead of added, and the correct solution is

$$x^2 y + \ln(xy) + e^y = constant\,.$$

Mark: 5 for correct solution, 2 for identifying error correctly